

LEARNING TO DISCRETIZE DENOISING DIFFUSION ODES

Vinh Tong, Trung-Dung Hoang, Anji Liu, Guy Van den Broeck, Mathias Niepert

University of Stuttgart, IMPRS-IS, UCLA, University of Bern

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Why fast diffusion sampling matters

- We want high-fidelity generations without paying hundreds of NFEs.
- At very small step counts, fixed or heuristic time schedules often break, driving truncation error and artifacts.
- We ask: can we learn solver- and model-adaptive time steps that preserve quality at low NFE without retraining the denoiser?
- Our goal: cut NFE while keeping quality by learning the discretization itself.

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LD3: learning to discretize diffusion ODEs

- We **learn** a strictly decreasing time grid for a chosen differentiable sampler and **differentiate through the sampler** to target global truncation error.
- We use **soft teacher forcing**: allow small input perturbations within an ϵ -ball for stable training, and we **show** this still **bounds divergence** from the teacher.
- We parameterize times with a **monotone cumulative-softmax**, optionally **decouple** solver and denoiser times, and **train in minutes** on 100 teacher pairs with rematerialization—**no denoiser retraining**.

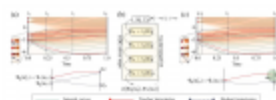


Figure: LD3 motivation and teacher-student scheduling: we minimize global truncation error with a soft objective.

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What we show and where it works

- We show large FID drops at low NFE on CIFAR10, AFHQv2, and FFHQ; for example, on CIFAR10 with iPNDM we reach FID 2.38 at 10 steps; on AFHQv2 we reach FID 2.27 at 10 steps.
- We evaluate on ImageNet and Stable Diffusion and see consistent improvements at small NFEs; beyond images, we improve point-cloud generation and molecular docking metrics at the same NFE.
- Practical notes: we train per solver and NFE; schedules are solver-specific; we do not retrain the denoiser.
- Limitations: requires differentiable solvers and training per NFE; 1–2 step regimes may still favor distillation.

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- Thank you for your attention!
- We welcome questions and feedback.